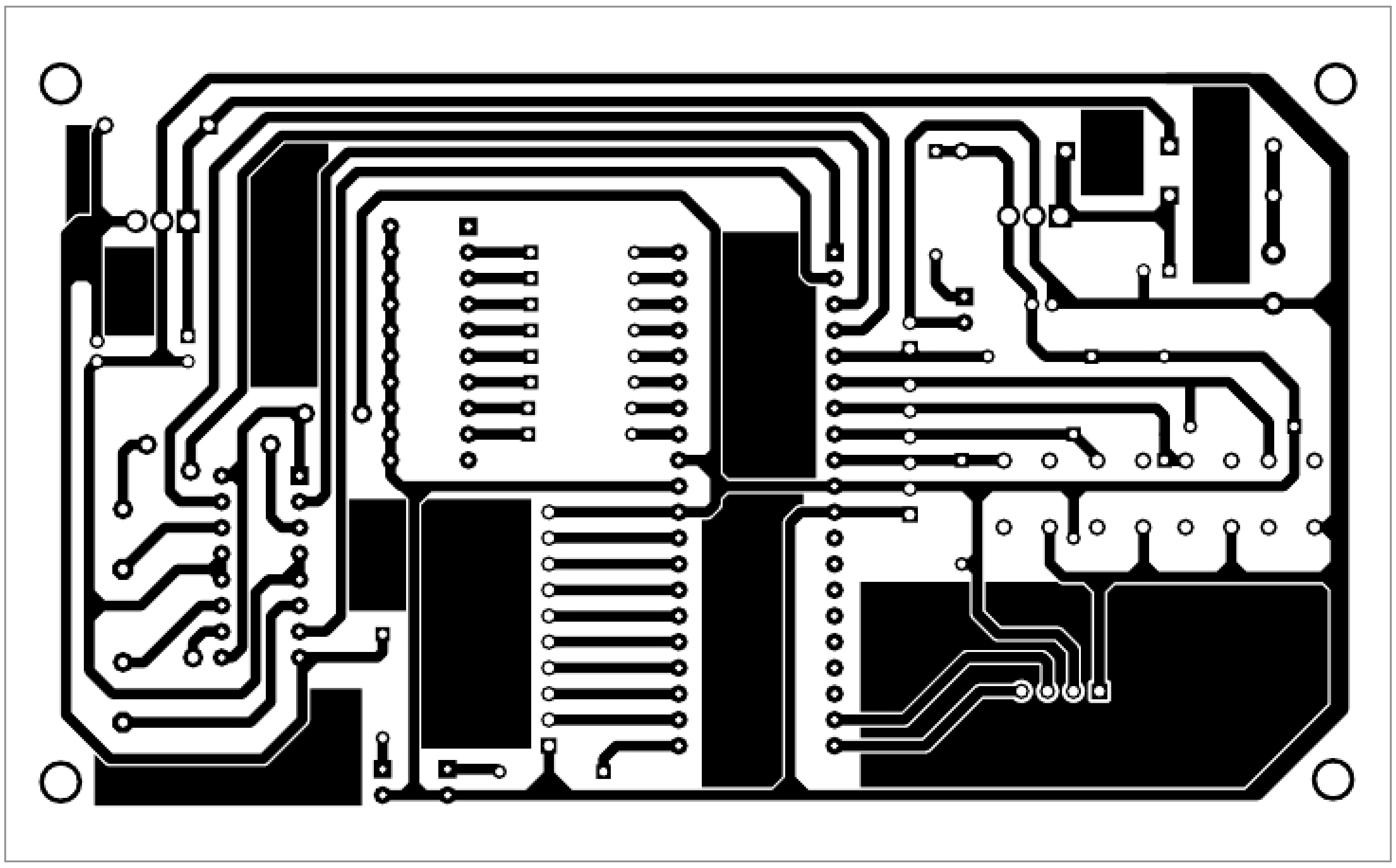

Fig. 4: Actual-size PCB layout for custom PID line follower

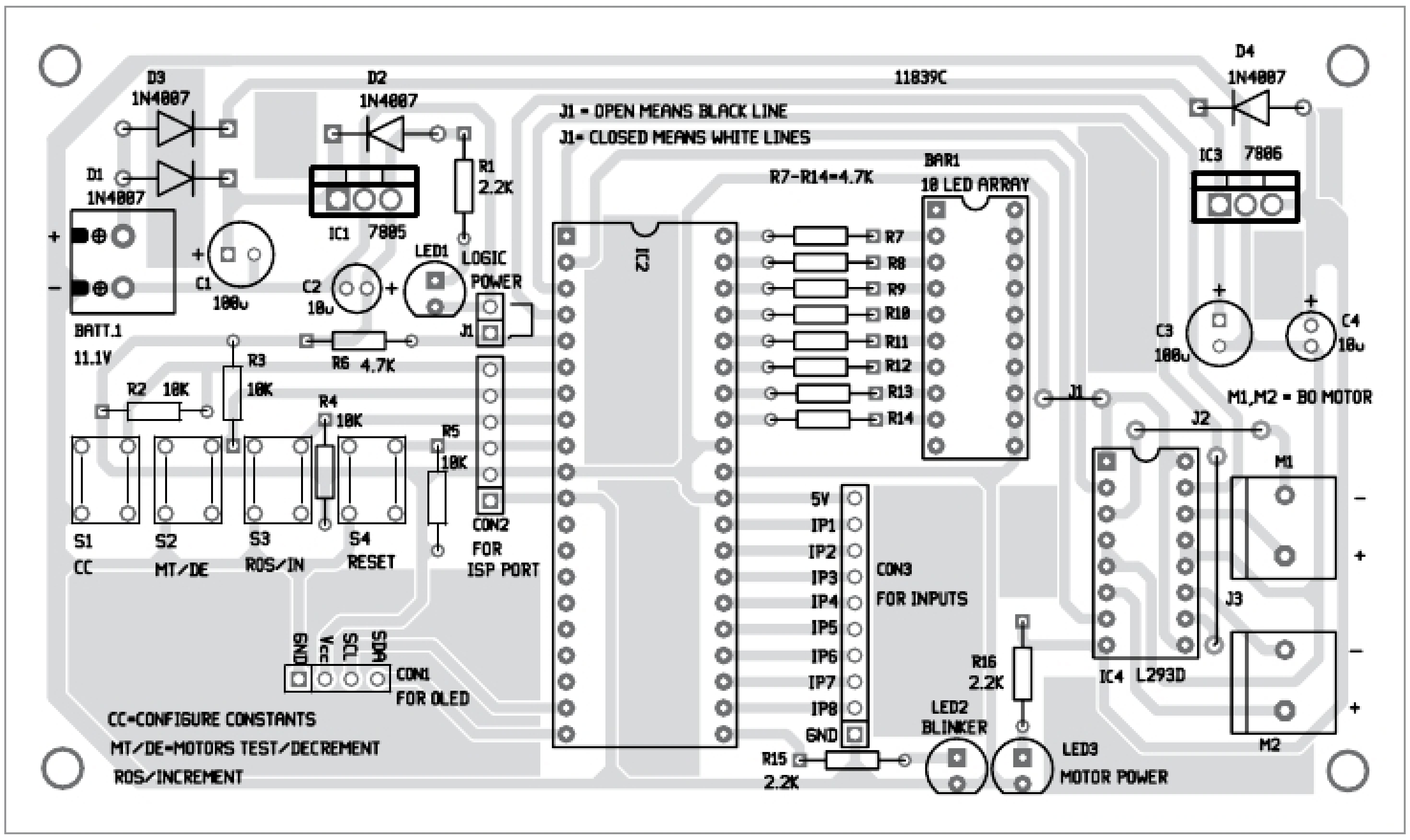

Fig. 5: Components layout of the PCB

off the power supply and remove the AVR programmer from the main circuit board.

Connect the sensor outputs to the main circuit unit and switch on the power supply of the sensor circuit unit. The LED bar array (BAR1) should blink, and the OLED should display 'welcome message' for a while, followed by the default 'stop' mode.

Now, adjust VR1 near op-amp ICs (LM324) in the sensor unit to turn on and off the eight LEDs in the LED bar array while moving the sensor unit on black (or white) line. Observe the difference of sensing the line by removing and connecting the shorting jumper J1 and set it as per the colour (black or white) of the line. Open J1 for black coloured line and close J1 for the white coloured line.

Self-test for motors

Press switch button S2 to test the rotation of the motors when the line follower robot is in Stop mode. The message on OLED would show 'Step 1 of 4, right motor is in forward mode.' At the same time, the right motor will move in a forward direction.

Next, press S2 momentarily to check the left motor. Similarly, press S2 to check the forward and backward movement of the line follower robot.

If any of the motors is not rotating, switch off the battery power supply and correct the wire connections of the motor. Then again go for a self-test and check until the directions of the motors match with the messages displayed on the OLED.

Configuring constants

For changing the constant values (Kp, Ki, and Kd), press button S1 successively to display current values and change the values.

Press button S2 to decrease or S3 to increase the values. If no button is pressed for more than five seconds, the changed values will be automatically saved to the EEPROM of the microcontroller.

The following parameters may be selected by pressing button S1 followed by pressing S2 or S3 buttons to increase or decrease the values, respectively:

1. Constant Kp (displayed as KP on OLED) is a multiplication constant for P in PID. You can change it to an appropriate value, say Kp = 8.1

2. Constant Ki (displayed as KI) is a multiplication constant for I in PID. Change it to say KI = 0.1

3. Constant Kd (displayed as KD) is a multiplication constant for D in PID. Change it to say Kd = 0.1

4. Loop delay (displayed as